

CYBER SECURITY COURSE

Redline Endpoint Investigation Method

Professional Case Report 05

Author	Balsa Klikovac
Report date	24. septembar 2026
Case ID	CSC-2026-05
Discipline	Incident response
Overall risk	High
Evidence status	Laboratory methodology documented

Purpose

Apply a repeatable host-triage sequence that can reveal persistence, unauthorized accounts and anomalous execution.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A Windows endpoint required a structured review of startup locations, accounts, files, event logs and active processes without relying on a single alert. A disciplined sequence produces stronger findings than ad hoc searching. The method is suitable as an initial corporate endpoint-response checklist.

Case Profile

Scenario	A Windows endpoint required a structured review of startup locations, accounts, files, event logs and active processes without relying on a single alert.
Environment	Windows workstation examined with native utilities and Redline-style evidence categories.
Tools	Redline, Autoruns, Registry Editor, WMIC, Tasklist, Windows Event Viewer

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Reviewed HKLM and HKCU startup locations and user startup folders.
2. Enumerated local users and Administrators group membership.
3. Searched for recent, hidden and unusually large files.
4. Compared running-process output from Task Manager, tasklist and WMIC.
5. Reviewed event logs for service changes, failed logons and security-control failures.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Multiple evidence sources required	Medium	No single Windows view provides sufficient context; process, registry, account and file evidence must be correlated.
F02	Startup locations are high-value	High	Registry Run keys, startup folders and scheduled actions remain common persistence points.
F03	Native tools support rapid triage	Low	WMIC, tasklist, net user and Event Viewer provide useful first-response coverage before specialist tooling is available.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
wmic startup get caption,command,user
net localgroup administrators
tasklist /v
wmic process list
```

Risk Assessment

Risk	Level	Operational impact
Missed persistence	High	Incomplete triage can leave an attacker active
Evidence alteration	Medium	Uncontrolled live-response actions may change volatile state

Recommendations

1. Use a documented collection order that starts with volatile evidence.
2. Record commands and timestamps during live response.
3. Compare suspicious files against a known-good baseline.
4. Escalate to full acquisition when indicators span multiple evidence categories.

Knowledge Demonstrated

- Endpoint triage
- Native Windows tooling
- Evidence correlation
- Investigation documentation

Conclusion

A disciplined sequence produces stronger findings than ad hoc searching. The method is suitable as an initial corporate endpoint-response checklist.

Hands On Lab Practical 05

Completed as part of the authorized cyber security course practical work.